GEOMETRY · FOUNDATIONS OF GEOMETRY

Distance & Midpoint Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Distance & Midpoint Explorer** on Mr. K's Classroom: Geometry → Foundations of Geometry → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. For A(1, 2) and B(4, 6), how could a right triangle help you find AB?

2. Where do you think the midpoint of (−2, 0) and (6, 0) is?

2 Explore use the tool and record what you see

Distance and midpoint

Drag A and B to each pair of points.

A	B	|Δx|	|Δy|	AB	Midpoint M
(0, 0)	(3, 4)				
(−4, −3)	(4, 3)				
(−3, 4)	(3, 4)				
(−4, −3)	(1, 2)				
(2, −5)	(−1, −1)				

Segment addition: A(−4, −3), B(1, 2)

Use the Segment addition tab and drag C.

Where C is	AC	CB	AC + CB
$\frac{1}{2}$ of the way			
$\frac{1}{5}$ of the way			
$\frac{3}{4}$ of the way			

3 Write the rule fill in each blank from what you observed

1. The distance formula comes from the _____: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

2. The coordinates of the midpoint are the _____ of the endpoints' coordinates.

3. If C is on segment AB, then AC + CB = _____.

4. C is the midpoint of AB exactly when AC _____ CB.

4 Check no tool: show your work

1. Find the distance from (–2, 1) to (4, 9).

2. Find the midpoint of (–2, 1) and (4, 9).

3. M(3, –1) is the midpoint of AB and A is (0, 2). Find B.

4. C is on AB with AC = x + 4, CB = 2x and AB = 19. Find x.

5 Reflect

Why does the distance formula square Δx and Δy?